Lecture 24

Berry's phase

Objectives. *A slowly cycled Hamiltonian can return an eigenstate with a measurable relative phase. Part of this phase depends on the elapsed time. Another part depends on the path of the parameters. We identify the second part with line-bundle holonomy. For the aligned spin-1/2 ground state, it is minus half the signed solid angle enclosed by the field-direction loop. The eigenvalue gap states where this description applies.*

By the end of this lecture you will be able to:

- State the gap and slow-evolution hypotheses of the cited adiabatic theorem.
- Construct the connection and curvature of a smooth simple eigenline.
- Compute a closed-cycle geometric phase with the correct gauge convention.
- Derive the spin-1/2 phase and its ground-state Chern number -1.

Unit 24A Eigenline transport and phase

24.1 Slow evolution and the gap

Recall the finite-dimensional quantum notation from *Quantum Mechanics I*. A pure state is a complex line in $\mathbb{C}^N$. A unit representative is written as a ket $|\psi\rangle$. Its conjugate transpose is the bra $\langle\psi|$, a linear functional. The inner product $\langle\phi|\psi\rangle = \phi^\dagger\psi$ is antilinear in its first argument. Multiplying a representative by a phase leaves its line unchanged.

Physical input. The Hamiltonian is a Hermitian matrix. A state vector evolves by the Schrödinger equation

$$\mathrm{i}\hbar\,\partial_t\,|\psi\rangle = H(t)\,|\psi\rangle\,.$$

Let $s \in [0,1]$ describe a fixed path of external parameters. Executing it in a duration $T > 0$ means $s = t/T$ for $0 \le t \le T$. Write the evolving vector as $\psi_T(s)$.

Definition 24.1 (Adiabatic limit)**.** The *adiabatic limit* is $T \to \infty$ with this path fixed.

Definition 24.2 (Simple eigenvalue and gap)**.** An eigenvalue is *simple* if its eigenspace has dimension one. Its *gap* is its distance to the other eigenvalues.

We first derive the transport equation by projecting a leading adiabatic ansatz. Choose a smooth unit eigenvector $|n(s)\rangle$ with eigenvalue $E_n(s)$ and write

$$\psi_{\mathrm{app}}(s) = D(s)c(s)\,|n(s)\rangle\,, \qquad D(s) = \exp\left(-\frac{\mathrm{i}T}{\hbar}\int_0^s E_n(u)\mathrm{d}u\right).$$

The terms containing TE_n cancel in the Schrödinger residual, leaving

$$i\hbar\psi'_{\text{app}} - TH\psi_{\text{app}} = i\hbar D\big(c'\,|n\rangle + c\,|n'\rangle\big).$$

Requiring the component of this residual along $|n\rangle$ to vanish gives the projected equation

$$c' + \langle n|n'\rangle\, c = 0.$$

This is Lecture 21's scalar parallel-transport equation. The duration T no longer occurs in it. The orthogonal part of the residual generally remains nonzero. This calculation motivates the connection but does not prove convergence of the full evolution.

Theorem 24.3 (Adiabatic evolution · *cited*). *Let $H(s)$ be a smooth family of Hermitian $N \times N$ matrices on $[0,1]$. Let $E_n(s)$ be a smooth simple eigenvalue separated from all other eigenvalues by a fixed $\delta > 0$. Start in a unit vector of its eigenspace and solve $i\hbar\,\psi_T'(s) = TH(s)\psi_T(s)$. There is a unique smooth unit vector $v(s)$ in that eigenspace with $v(0) = \psi_T(0)$ and $\langle v(s)|v'(s)\rangle = 0$. For each fixed $s \in [0,1]$, the cited theorem gives*

$$\left\| \psi_T(s) - \exp\left(-\frac{iT}{\hbar}\int_0^s E_n(u)\,du\right) v(s) \right\| \longrightarrow 0 \quad \text{as } T \to \infty.$$

This is the finite-dimensional form of Kato's adiabatic theorem. See Barry Simon, p. 2167, for the gap assumption and phase separation, and p. 2168 for the transport condition on the limit.

Definition 24.4 (Dynamical phase). The exponential in Theorem 24.3 is the *dynamical phase factor*.

Definition 24.5 (Geometric phase). If $H(1) = H(0)$ and the chosen eigenline returns to itself, then $v(1) = e^{i\gamma(C)}v(0)$ for a phase defined modulo 2π. This scalar is the *geometric phase factor*, or *Berry phase factor*.

The symbol C denotes the parameter cycle, whereas $\gamma(C)$ denotes its phase.

The theorem is a limit statement. A changing eigenvector need not solve the full Schrödinger equation exactly at finite T.

The figure separates a gap condition from the choice of eigenvectors.

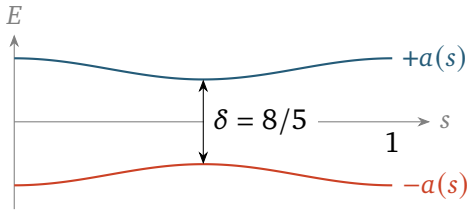


Figure 24.1: The diagonal family $H(s) = \operatorname{diag}(-a(s), a(s))$, with $a(s) = 1 + \cos(2\pi s)/5$, has a gap at least $8/5$. Its eigenspaces are constant, so its geometric phase is zero.

A varying eigenline occurs in the avoided-crossing family

$$H(s) = \begin{pmatrix} \Lambda(s-\frac{1}{2}) & \varepsilon \\ \varepsilon & -\Lambda(s-\frac{1}{2}) \end{pmatrix}, \qquad \Lambda > 0, \quad \varepsilon > 0,$$

$$E_\pm(s) = \pm\sqrt{\Lambda^2(s-\tfrac{1}{2})^2 + \varepsilon^2}.$$

Squaring the Hamiltonian gives the displayed energies. The minimum gap is $\delta = 2\varepsilon$ at $s = 1/2$. The lower eigenline varies along the path. The figure supplements the preceding constant-eigenspace family.

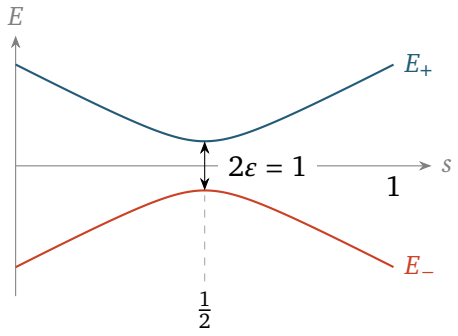


Figure 24.2: The exact levels for $\Lambda = 4$ and $\varepsilon = 1/2$ have minimum gap 1. The eigenlines vary even though the two levels never meet.

A heuristic scale for slow evolution is

$$\frac{\hbar \max_{s\in[0,1]} \|H'(s)\|}{T\delta^2} \ll 1.$$

To see the two gap factors, differentiate $H|n\rangle = E_n|n\rangle$ and project onto an orthogonal unit eigenvector $|r\rangle$ with eigenvalue E_r. The result is

$$\langle r|n'\rangle = \frac{\langle r|H'|n\rangle}{E_n - E_r}.$$

The relative dynamical oscillation has frequency $T(E_r - E_n)/\hbar$ in s. Together these suggest the displayed scale. For the avoided crossing, $\|H'\| = \Lambda$, so it reads $T \gg \hbar\Lambda/(4\varepsilon^2)$. This is a scale estimate, not a quantitative error theorem. Such a theorem also needs bounds on further derivatives and the endpoint behavior of the family. A zero gap invalidates the cited theorem's hypothesis. It does not rule out every exact eigenvector evolution through a crossing.

24.2 Eigenlines and their connection

Let $\mathcal{U}$ be an open region in a smooth parameter manifold, with a smooth family $H\colon \mathcal{U} \to \mathrm{Herm}(N)$ and a chosen smooth simple eigenvalue $E_n(R)$. Here $\mathrm{Herm}(N)$ is the real vector space of Hermitian matrices.

Definition 24.6 (Eigenline bundle). The *eigenline bundle* is the subset of $\mathcal{U} \times \mathbb{C}^N$ given by

$$L_n = \{(R, w) : R \in \mathcal{U},\ H(R)w = E_n(R)w\}.$$

Its fibre over R is the eigenspace, including its zero vector.

The index n labels the chosen energy branch. The integer N is the dimension of the ambient state space. Smoothness of the selected branch is an explicit hypothesis here. It holds for the separated branches used below. Along a smooth parameter path $R(s)$, the adiabatic theorem applies to $H(R(s))$ when its eigenvalue has the stated uniform gap.

Definition 24.7 (Berry connection)**.** The *Berry connection* of the eigenline bundle is the projected derivative $\nabla\xi = P\mathrm{d}\xi$, where $P(R)$ is the orthogonal projection onto the eigenline.

Lecture 16 constructed a surface connection by differentiating in $\mathbb{R}^3$ and projecting onto the tangent plane. Here we differentiate in $\mathbb{C}^N$ and project onto the eigenline. The figure compares these projections on exact real curves.

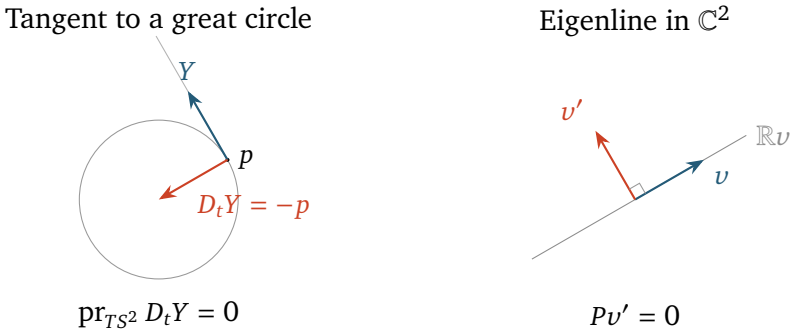


Figure 24.3: Along the unit sphere's great circle $p = (\cos t, \sin t, 0)$, the field $Y = p'$ has ambient derivative $Y' = -p$ and zero tangent projection. For the real unit vector $v = (\cos t, \sin t)$ in $\mathbb{C}^2$, the derivative is orthogonal to its complex line, so $Pv' = 0$.

Theorem 24.8 (The Berry connection · *proved geometrically; adiabatic interpretation given the cited theorem*)**.** *The eigenlines form a smooth Hermitian line bundle. For a smooth unit eigenvector $|n_i\rangle$ on $U_i \subset \mathcal{U}$, the* Berry connection *has imaginary local one-form*

$$\mathcal{A}_i = \langle n_i | \mathrm{d}n_i \rangle .$$

These forms define the projected derivative $\nabla\xi = P\mathrm{d}\xi$, where $P(R)$ is the orthogonal projection onto the eigenline. Its parallel transport is the vector $v(s)$ in the adiabatic theorem. Thus its holonomy around a cycle C is $e^{\mathrm{i}\gamma(C)}$.

Proof. The matrix $H(R) - E_n(R)I$ has rank $N-1$. Its adjugate, the transposed cofactor matrix, has a nonzero column at any fixed point. Its columns lie in the kernel of the original matrix, by the identity $B\,\mathrm{adj}(B) = \det(B)I$ for every square matrix B. Choose such a column. It stays nonzero nearby and is smooth in R. Divide it by its positive norm to obtain a smooth unit eigenvector $|n_i\rangle$. Its scalar multiples give a local product chart for L_n. On overlaps write $|n_j\rangle = |n_i\rangle\, g_{ij}$, where $g_{ij} = \langle n_i | n_j \rangle \in \mathrm{U}(1)$. These smooth transitions satisfy the cocycle law, so Lecture 20's charts give the bundle.

The projection is $P = |n_i\rangle\langle n_i|$ and is unchanged by this unit phase. It is therefore global. For a section $\xi = |n_i\rangle\, c$, direct differentiation gives

$$P\mathrm{d}\xi = |n_i\rangle \left(\mathrm{d}c + \langle n_i | \mathrm{d}n_i \rangle\, c\right).$$

The ordinary product rule gives the connection Leibniz rule because $P\xi = \xi$. Self-adjointness of P similarly gives the Hermitian metric product rule. Differentiating $\langle n_i | n_i \rangle = 1$ shows $\mathcal{A}_i + \overline{\mathcal{A}_i} = 0$, so the potential is imaginary. Finally, for a vector $v(s)$ in the eigenline, its projection of $v'(s)$ is $v(s)\,\langle v(s) | v'(s) \rangle$ when v is unit. Thus the adiabatic theorem's condition is exactly $\nabla_{\partial_s} v = 0$. Uniqueness of Lecture 21's transport identifies the two vectors and their endpoint phase. ∎

The bundle and connection depend only on the eigenspaces. The dynamical phase also uses the eigenvalues and traversal time.

24.3 Curvature, gauge changes and holonomy

A different unit vector on one chart has the form $|n'\rangle = |n\rangle\, h$, with $h\colon U \to \mathrm{U}(1)$. Differentiating the product and using unit norm gives

$$\mathcal{A}' = \mathcal{A} + h^{-1}\mathrm{d}h, \qquad \mathcal{A}' = \mathcal{A} + \mathrm{i}\mathrm{d}\chi \quad \text{if } h = e^{\mathrm{i}\chi}.$$

This is Lecture 21's passive line-frame law. The coefficient of a fixed section changes by $c' = h^{-1}c$.

Definition 24.9 (Real Berry potential)**.** The *real Berry potential* is $A_{\mathrm{B}} = \mathrm{i}\mathcal{A}$.

Definition 24.10 (Berry curvature)**.** The *Berry curvature* is the global two-form $\mathrm{d}\mathcal{A}$.

The real potential differs from the imaginary connection form used in the transport equation. Lecture 18 writes $\mathrm{d}A$ and Lecture 22 writes $\mathrm{d}\mathcal{A}$ for a surface area form. Neither denotes the Berry connection $\mathcal{A}$ or its curvature $\mathrm{d}\mathcal{A}$ used here. Differentiating the local connection yields the full local formula

$$\mathrm{d}\mathcal{A} = \langle \mathrm{d}n| \wedge |\mathrm{d}n\rangle = \sum_{a=1}^{N} \mathrm{d}\overline{n^a} \wedge \mathrm{d}n^a.$$

Here the bra–ket wedge denotes the componentwise sum on the right. The term with $\mathrm{d}^2 n$ vanishes. The gauge law shows that the curvature is unchanged on overlaps. Thus Lecture 22 supplies the real class $c_1(L_n) = [\mathrm{i}\mathrm{d}\mathcal{A}/(2\pi)]$.

In a periodic unit frame along a closed cycle, the scalar transport equation gives

$$e^{\mathrm{i}\gamma(C)} = \exp\left(-\oint_C \mathcal{A}\right), \qquad \gamma(C) = \mathrm{i}\oint_C \mathcal{A} = \oint_C A_{\mathrm{B}} \pmod{2\pi}.$$

Such a frame exists along the cycle even when none exists on all of $\mathcal{U}$. Indeed, transport a unit vector along the parameter interval to get $v(s)$. If $v(1) = e^{\mathrm{i}\alpha}v(0)$, the vector $e^{-\mathrm{i}\alpha s}v(s)$ is periodic. Alternatively, use local frames and the transition factors when changing charts, as in Lecture 21. Under any single-valued gauge map on the cycle, $\oint_C h^{-1}\mathrm{d}h = 2\pi\mathrm{i}\,\mathrm{wind}(h)$. The geometric phase changes by an integer multiple of 2π, leaving its factor invariant. No global real logarithm of h is required. A parallel unit vector can therefore have zero connection coefficient along the interval without being periodic at its endpoints. A periodic change of frame cannot remove a nontrivial closed-cycle holonomy.

If one frame covers an oriented compact spanning surface Σ with $\partial\Sigma = C$, Stokes gives $\gamma(C) = \mathrm{i}\int_\Sigma \mathrm{d}\mathcal{A}$ modulo 2π. The surface and frame hypotheses matter, as they did for the Aharonov–Bohm calculation. The real potential also gives the local phase convention used in Lecture 21: A_{B} corresponds to $(q/\hbar)A_{\mathrm{em}}$. This compares connections on different parameter spaces, rather than identifying the physical fields.[1]

Physical input. In an ideal coherent two-arm interferometer, the complex amplitudes of the arms add at recombination. The detection intensity is proportional to the squared modulus

[1] Berry's equation (6) uses our $\mathrm{i}\oint \mathcal{A}$. Tong uses $A_{\mathrm{Tong}} = -\mathrm{i}\mathcal{A} = -A_{\mathrm{B}}$. His factor $\exp(-\mathrm{i}\oint A_{\mathrm{Tong}})$ is the same holonomy.

of their sum. We assume that both arms end in the same internal eigenline with the same normalized representative.

We cycle the parameters in one arm and hold them fixed in the reference arm. If their energy histories have equal integrals and their durations agree, their dynamical phases cancel in the relative phase. For arm amplitudes a_{cyc} and a_{ref}, the detection intensity is proportional to

$$|a_{\mathrm{cyc}}e^{\mathrm{i}\gamma(C)} + a_{\mathrm{ref}}|^2 = |a_{\mathrm{cyc}}|^2 + |a_{\mathrm{ref}}|^2 + 2\,\mathrm{Re}\big(\overline{a_{\mathrm{ref}}}a_{\mathrm{cyc}}e^{\mathrm{i}\gamma(C)}\big).$$

Thus a phase of one arm changes interference with the other arm. For a constant energy $-b$, the dynamical phase angle is $bT/\hbar$. Doubling T doubles this angle while leaving the limiting geometric phase fixed. Reversing the loop with the same duration and the same energy history in reverse order leaves the dynamical integral unchanged. It reverses the geometric phase. Comparing these two traversals gives $2\gamma(C)$ modulo 2π. This comparison alone determines $\gamma(C)$ only modulo π. The figure shows the reference-arm scheme for the spin rotation computed in §24.4.

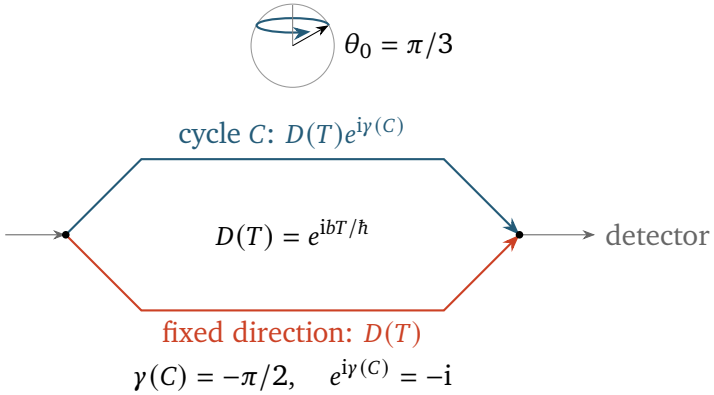


Figure 24.4: Both arms have energy $-b$ and duration T, so their common dynamical factor cancels. For the illustrated field-direction latitude $\theta_0 = \pi/3$, the cycled arm acquires the geometric factor $-\mathrm{i}$ in the adiabatic limit.

Self-check. State in one sentence what the Berry connection is and why it was introduced.

Answer. The Berry connection is the projected derivative $P\mathrm{d}$ on the eigenline bundle, introduced because the cited adiabatic theorem identifies its parallel transport with evolution after removal of the dynamical phase. Indeed, $P\mathrm{d}(|n\rangle\, c) = |n\rangle\,(\mathrm{d}c + \langle n|\mathrm{d}n\rangle\, c)$ gives the projected Schrödinger equation.

Self-check. For a smooth open path $C\colon [0,1] \to \mathcal{U}$, explain how $\mathrm{i}\int_C \mathcal{A}$ changes under $|n\rangle \mapsto e^{\mathrm{i}\chi}\,|n\rangle$, for a single-valued real χ on the path.

Answer. The gauge law gives

$$\mathrm{i}\int_C (\mathcal{A} + \mathrm{i}\mathrm{d}\chi) = \mathrm{i}\int_C \mathcal{A} - [\chi(C(1)) - \chi(C(0))].$$

Thus this integral alone is not an intrinsic open-path phase. An endpoint reference must also transform if an open-path interference measurement is specified. For a closed cycle the holonomy is intrinsic and its phase is defined modulo 2π.

Unit 24B Spin eigenlines and solid angle

24.4 A spin-$1/2$ in a rotating field

Recall the Pauli matrices and the spin operators S_a, for $a \in \{x, y, z\}$:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad S_a = \frac{\hbar}{2}\sigma_a.$$

The symbol σ_a denotes these matrices, distinct from Lecture 21's section-coefficient column. Their products obey $\sigma_a\sigma_b = \delta_{ab}I + \mathrm{i}\epsilon_{abc}\sigma_c$. Use an energy-scaled field $\mathbf{B} = b\mathbf{n}$, with $b > 0$ and $|\mathbf{n}| = 1$.

Physical input. We model a spin-1/2 in the energy-scaled field $\mathbf{B}$ by the Hamiltonian $H = -\mathbf{B} \cdot \boldsymbol{\sigma}$.

Its eigenvalues are $-b$ and $+b$ because $(\mathbf{n}\cdot\boldsymbol{\sigma})^2 = I$. Its gap is $2b$ everywhere on the direction sphere. The earlier avoided-crossing matrix is also of this Pauli form: $H(s) = \Lambda(s - \frac{1}{2})\sigma_z + \varepsilon\sigma_x$.

For a smooth loop $C\colon [0,1] \to S^2$ based at $p = C(0) = C(1)$, choose a continuous filling $f\colon [0,1]^2 \to S^2$, smooth on each rectangle of a finite subdivision, with

$$f(s,0) = C(s), \qquad f(s,1) = f(0,u) = f(1,u) = p.$$

Lecture 20's hemisphere, polygon and stereographic construction supplies such a filling, smooth on each rectangle of a finite subdivision. Orient the square by $\mathrm{d}s \wedge \mathrm{d}u$. Recall the outward unit-sphere area form β from Lecture 12.

Definition 24.11 (Signed solid angle). The *signed solid angle* of this filling is

$$\Omega_f(C) = \int_{[0,1]^2} f^*\beta.$$

The proof of Proposition 24.12 shows that its class modulo 4π is independent of the filling. We denote this class by $\Omega(C)$. For a simple loop bounding an outward-oriented region D on its left, one filling has signed area $\mathrm{area}(D)$. The complementary region must have its orientation reversed to have the same boundary. Its signed area is $\mathrm{area}(D) - 4\pi$. The resulting phases differ by 2π and give the same factor. The figure shows this ambiguity for a latitude.



$\gamma = -\pi/2$ $\qquad$ $\gamma = 3\pi/2$

Figure 24.5: At $\theta_0 = \pi/3$, the northern cap has signed area π and the reversed complementary cap has signed area -3π. Both have the same oriented boundary and give geometric factor $-\mathrm{i}$.

Proposition 24.12 (The spin eigenline · *proved*)**.** *For* $\mathbf{n} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$*, the aligned ground-state eigenline is Lecture 20's tautological line on* $\mathbb{C}\mathrm{P}^1$*. On the northern chart it has*

$$|n_N\rangle = \begin{pmatrix} \cos(\theta/2) \\ e^{\mathrm{i}\phi}\sin(\theta/2) \end{pmatrix}, \quad \mathcal{A}_N = \frac{\mathrm{i}}{2}(1-\cos\theta)\mathrm{d}\phi, \quad \mathrm{d}\mathcal{A} = \frac{\mathrm{i}}{2}\sin\theta\mathrm{d}\theta\wedge\mathrm{d}\phi.$$

In the outward orientation its Chern number is -1*. For an oriented loop, the geometric phase is* $\gamma(C) = -\Omega(C)/2$ *modulo* 2π*, where the solid angle is defined modulo* 4π*.*

Proof. The direction matrix is

$$\mathbf{n}\cdot\boldsymbol{\sigma} = \begin{pmatrix} \cos\theta & e^{-\mathrm{i}\phi}\sin\theta \\ e^{\mathrm{i}\phi}\sin\theta & -\cos\theta \end{pmatrix}.$$

Multiplying by the displayed vector gives $|n_N\rangle$, so $H|n_N\rangle = -b|n_N\rangle$. Its norm is one. In $z = \tan(\theta/2)e^{\mathrm{i}\phi}$ it is $(1,z)^T/\sqrt{1+|z|^2}$, the tautological unit frame of Lecture 20. Its Pauli expectations are

$$\langle n_N|\sigma_x|n_N\rangle = \sin\theta\cos\phi, \qquad \langle n_N|\sigma_y|n_N\rangle = \sin\theta\sin\phi, \qquad \langle n_N|\sigma_z|n_N\rangle = \cos\theta.$$

Thus $\mathbf{n}$ is the state's Bloch vector. The unit eigenvectors form Lecture 20's Hopf circle bundle $S^3 \to S^2$. The inner product with its θ derivative is $-\cos(\theta/2)\sin(\theta/2)/2 + \sin(\theta/2)\cos(\theta/2)/2 = 0$. The ϕ derivative gives $\mathrm{i}\sin^2(\theta/2) = \mathrm{i}(1-\cos\theta)/2$. Differentiating gives the curvature stated above.

The southern unit vector is $|n_S\rangle = e^{-\mathrm{i}\phi}|n_N\rangle$. It is smooth at the south pole and has $\mathcal{A}_S = \mathcal{A}_N - \mathrm{i}\mathrm{d}\phi$. Thus its frame transition has winding -1. These are exactly Lecture 22's Wu–Yang potentials with monopole integer $n = -1$. The eigenline is its degree--1 line $O(-1)$. Here the subscripts N and S name the northern and southern charts. Equivalently, direct curvature integration gives

$$\frac{\mathrm{i}}{2\pi}\int_{S^2}\mathrm{d}\mathcal{A} = -\frac{1}{4\pi}\int_0^{2\pi}\int_0^{\pi}\sin\theta\mathrm{d}\theta\mathrm{d}\phi = -1.$$

The nonzero Chern number prevents a smooth unit eigenvector on the whole sphere. To compute the phase for a general loop, we instead pull the eigenline and connection back to the filling square. We construct a unit frame $w(s,u)$ by vertical transport from a fixed vector on the top edge. The figure shows this construction and the boundary orientation used in Stokes.

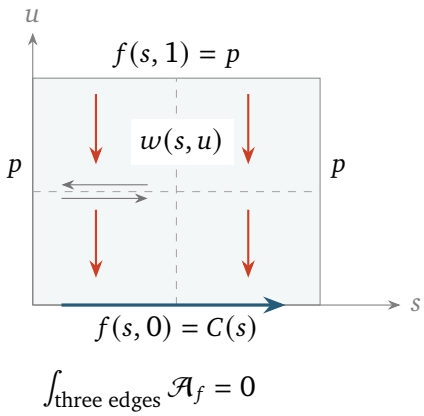


Figure 24.6: The bottom edge runs in increasing s for the orientation $\mathrm{d}s \wedge \mathrm{d}u$. Vertical transport from the fixed unit vector on the top edge makes $w(s,u)$ constant on the three edges mapped to p, so only the bottom integral survives after shared edges cancel.

Fix a unit vector over p. From the top edge, transport it down each vertical segment. Smooth dependence of transport on parameters gives a unit frame $w(s,u)$, continuous on the square and smooth on each rectangle of the finite subdivision. The top and side edges map constantly to p, so the frame is the same fixed vector on all three. In particular, its bottom restriction is a periodic frame along C.

Let $\mathcal{A}_f = \langle w|\mathrm{d}w\rangle$ be its connection form on the square. On each rectangle, its curvature is $\mathrm{d}\mathcal{A}_f = f^*\mathrm{d}\mathcal{A}$. The tangential restrictions on shared edges agree, so internal boundary integrals cancel. The top and side integrals vanish because their frame is constant. The positively oriented bottom edge runs in increasing s. Stokes on the rectangles therefore gives

$$\gamma(C) = \mathrm{i}\int_0^1 \mathcal{A}_f(\partial_s)(s,0)\mathrm{d}s = \mathrm{i}\int_{[0,1]^2} f^*\mathrm{d}\mathcal{A} = -\frac{1}{2}\Omega_f(C) \pmod{2\pi}.$$

Two fillings f and g compute the same intrinsic holonomy. Hence $\exp[-\mathrm{i}(\Omega_f(C)-\Omega_g(C))/2] = 1$, so their signed areas differ by an integer multiple of 4π. This proves both the solid-angle convention and the claimed phase formula. ∎

The figure fixes the orientation of a latitude and its northern cap.

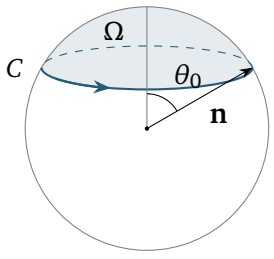


Figure 24.7: Increasing azimuth orients the latitude as the boundary of the northern cap of solid angle $\Omega = 2\pi(1-\cos\theta_0)$. Its spin ground-state geometric phase is $-\Omega/2$.

Example 24.13 (A closed field rotation)**.** Let $b > 0$ be fixed, choose $0 < \theta_0 < \pi$, and rotate the field with $\phi(t) = 2\pi t/T$ for $0 \le t \le T$. Start in $|n_N(\theta_0, 0)\rangle$ and take the adiabatic limit.

We use $H = -b\mathbf{n}\cdot\boldsymbol{\sigma}$. On the northern cap, the ground line is spanned by

$$|n_N(\theta,\phi)\rangle = \begin{pmatrix} \cos(\theta/2) \\ e^{\mathrm{i}\phi}\sin(\theta/2) \end{pmatrix}.$$

Differentiating this unit vector gives its connection

$$\mathcal{A}_N = \langle n_N|\mathrm{d}n_N\rangle = \mathrm{i}\sin^2(\theta/2)\mathrm{d}\phi = \frac{\mathrm{i}}{2}(1-\cos\theta)\mathrm{d}\phi.$$

The northern cap $D = \{0 \le \theta \le \theta_0\}$ carries this frame and has boundary C in increasing ϕ. Stokes on this cap gives

$$\mathrm{i}\int_D \mathrm{d}\mathcal{A}_N = -\frac{1}{2}\int_0^{2\pi}\int_0^{\theta_0}\sin\theta\mathrm{d}\theta\mathrm{d}\phi = -\pi(1-\cos\theta_0) = -\frac{1}{2}\Omega(C).$$

The northern unit frame is periodic on this path, so direct transport gives the same value:

$$\gamma(C) = \mathrm{i}\int_0^{2\pi}\frac{\mathrm{i}}{2}(1-\cos\theta_0)\mathrm{d}\phi = -\pi(1-\cos\theta_0).$$

The energy is $-b$, so the dynamical factor is $e^{+\mathrm{i}bT/\hbar}$. The final state is asymptotic to

$$e^{+\mathrm{i}bT/\hbar}e^{-\mathrm{i}\pi(1-\cos\theta_0)}\,|n_N(\theta_0,0)\rangle\,.$$

For $\theta_0 = \pi/3$ the geometric factor is $-\mathrm{i}$. The southern potential instead gives a phase larger by 2π, hence the same factor.

The orthogonal excited vector $(-e^{-\mathrm{i}\phi}\sin(\theta/2), \cos(\theta/2))^T$ has $\mathcal{A}_{\mathrm{exc}} = -\mathrm{i}(1-\cos\theta)\mathrm{d}\phi/2$. Its curvature and Chern number have the opposite sign, giving $+1$.

Using the same cap, Lecture 18's tangent-vector transport angle is Ω. The spin phase is its negative half before either angle is reduced modulo 2π. At the equator the vector holonomy is a full turn while the spin phase factor is -1.

In Lecture 18's smooth northern tangent frame, the connection entry is $\omega_N = (1-\cos\theta)\mathrm{d}\phi$. The complex tangent-line connection is $-\mathrm{i}\omega_N$, whereas $\mathcal{A}_N = \mathrm{i}\omega_N/2$. In these frames the aligned spin-line connection is minus half the tangent-line connection. For $\theta_0 = \pi/3$, parallel transport through azimuth ϕ has spin phase angle $-\phi/4$ and tangent-vector angle $\phi/2$ relative to these northern frames. The figure samples both angles along the same latitude.

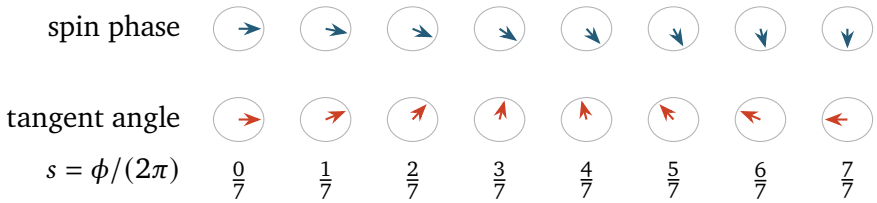


Figure 24.8: The eight samples have $\phi = 2\pi k/7$ for $k = 0,\dots,7$ and $\theta_0 = \pi/3$. The spin phase ends at $-\pi/2$, while the tangent vector ends at angle $+\pi$ in its northern frame.

The spin coefficient also describes a lift into the phase circles over the parameter loop. The figure opens the loop at its base point to display both endpoint representatives.

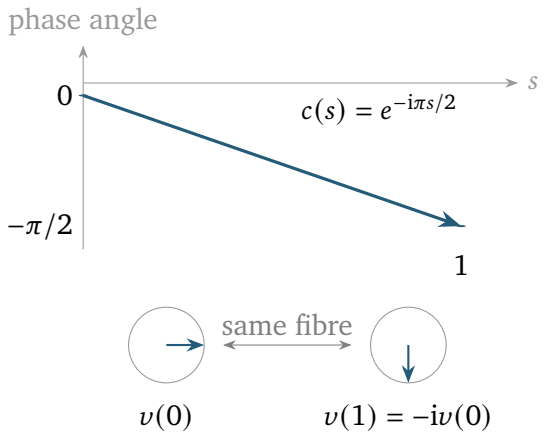


Figure 24.9: For the same latitude, $c(s) = e^{-i\pi s/2}$ gives the parallel vector $v(s) = c(s)\,|n_N(s)\rangle$. The endpoint circles represent the same fibre, but the parallel vector returns multiplied by $-i$.

24.5 The degeneracy and general spin

For $\mathbf{B} \neq 0$, the ground-state projector is $P(\mathbf{B}) = (I + \mathbf{B}\cdot\boldsymbol{\sigma}/|\mathbf{B}|)/2$. It depends only on the field direction. Pulling the sphere curvature back to $\mathbb{R}^3 \setminus \{0\}$ gives

$$\mathrm{d}\mathcal{A} = \frac{\mathrm{i}}{2|\mathbf{B}|^3}\left(B^1\mathrm{d}B^2 \wedge \mathrm{d}B^3 + B^2\mathrm{d}B^3 \wedge \mathrm{d}B^1 + B^3\mathrm{d}B^1 \wedge \mathrm{d}B^2\right).$$

The real curvature $\mathrm{d}A_{\mathrm{B}} = \mathrm{i}\mathrm{d}\mathcal{A}$ points inward. Its flux through an outward-oriented sphere is -2π. This is Lecture 22's $n = -1$ line in field-parameter space. The figure shows its inward real Berry curvature.

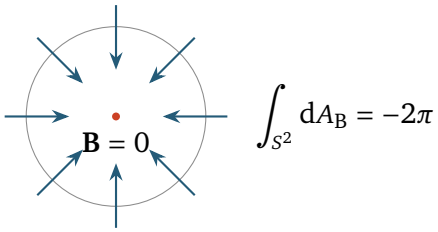


Figure 24.10: The real Berry curvature points inward in field-parameter space. Any outward sphere around the excluded degeneracy has $\int \mathrm{d}A_{\mathrm{B}} = -2\pi$ and ground-line Chern number -1: this is Lecture 22's $n = -1$ line.

At $\mathbf{B} = 0$ the two eigenvalues coincide. Every line is then an eigenline, so the selected ground line and its projector are not determined. For the identically zero Hamiltonian the exact evolution is constant. It does not force a prescribed changing eigenline to be followed. The gap hypothesis prevents this ambiguity.

A nonzero Chern number on the boundary of a parameter ball forces a degeneracy inside if the Hamiltonian and the selected eigenvalue branch are smooth throughout the ball. Indeed, suppose the branch stayed simple there. Radial transport from the centre would give a smooth unit frame on the ball, as in Lecture 22. Its connection form $\mathcal{A}$ would then

be global, so Stokes would give

$$C_1(\partial B^3) = \frac{\mathrm{i}}{2\pi}\int_{\partial B^3} \mathrm{d}\mathcal{A} = \frac{\mathrm{i}}{2\pi}\int_{B^3} \mathrm{d}^2\mathcal{A} = 0.$$

This contradicts a nonzero boundary Chern number. The figure compares a sphere enclosing the degeneracy with one bounding a ball that avoids it.

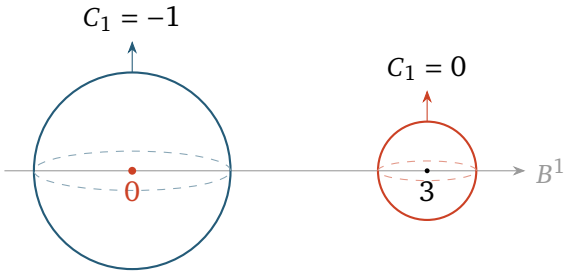


Figure 24.11: For $H = -\mathbf{B}\cdot\boldsymbol{\sigma}$, the unit sphere about the origin has ground-line Chern number -1. The sphere of radius $1/2$ about $(3,0,0)$ bounds a ball avoiding the origin and has Chern number zero.

Every Hermitian 2×2 matrix has the form

$$H = E_0 I + \mathbf{d}\cdot\boldsymbol{\sigma} = \begin{pmatrix} E_0 + d_z & d_x - \mathrm{i}d_y \\ d_x + \mathrm{i}d_y & E_0 - d_z \end{pmatrix}, \qquad E_0, d_x, d_y, d_z \in \mathbb{R}.$$

Its levels coincide exactly when $\mathbf{d} = 0$. For a smooth three-parameter family, assume that the derivative of $\mathbf{d}$ at such a zero is invertible. The inverse function theorem then makes $\mathbf{d}$ a local coordinate system. An enclosing small sphere has lower-line Chern number $+1$ or -1 according to the orientation of these coordinates. Indeed, its lower line pulls back the excited line of $-\mathbf{n}\cdot\boldsymbol{\sigma}$, whose Chern number is $+1$. Lecture 14's degree formula multiplies this by the sign of the coordinate Jacobian. The field model here has $\mathbf{d} = -\mathbf{B}$, which reverses orientation and gives -1. For a larger Hamiltonian this description applies to a separated smooth two-level block under the same invertibility hypothesis. Crossings constrained by symmetries or with singular derivative need not have this local form.

For general spin, we assume the following additional matrix data.

Proposition 24.14 (Spin j · *sketched*)**.** *Let $j > 0$ be an integer or half-integer. Assume Hermitian dimensionless spin matrices $J_a = S_a/\hbar$ on $\mathbb{C}^{2j+1}$, with $[J_a, J_b] = \mathrm{i}\epsilon_{abc}J_c$. Assume an orthonormal basis $|j,\ell\rangle$ with*

$$J_z\,|j,\ell\rangle = \ell\,|j,\ell\rangle\,, \qquad \ell \in \{-j, -j+1, \dots, j\}.$$

On the outward direction sphere, for the eigenline of $\mathbf{n}\cdot\mathbf{J}$ of weight $m \in \{-j, -j+1, \dots, j\}$,

$$\mathcal{A}_N = \mathrm{i}m(1-\cos\theta)\mathrm{d}\phi, \qquad \gamma(C) = -m\Omega(C) \pmod{2\pi}, \qquad C_1 = -2m.$$

Sketch. Choose $J_z\,|j,m\rangle = m\,|j,m\rangle$ and rotate the weight vector by

$$|n_N\rangle = e^{\mathrm{i}m\phi}e^{-\mathrm{i}\phi J_z}e^{-\mathrm{i}\theta J_y}\,|j,m\rangle\,.$$

The combined full-turn operator acts on every basis vector by

$$e^{2\pi\mathrm{i}m}e^{-2\pi\mathrm{i}J_z}\,|j,\ell\rangle = e^{2\pi\mathrm{i}(m-\ell)}\,|j,\ell\rangle = |j,\ell\rangle\,,$$

because $m - \ell \in \mathbb{Z}$. Thus the rotated vector is single-valued in ϕ. Differentiating the conjugated operators and using the commutators gives $e^{i\theta J_y} J_z e^{-i\theta J_y} = J_z \cos\theta - J_x \sin\theta$. In a J_z weight vector the expectation of any commutator with J_z is zero. The relations $[J_z, J_y] = -iJ_x$ and $[J_z, J_x] = iJ_y$ therefore give $\langle j, m| J_x |j, m\rangle = \langle j, m| J_y |j, m\rangle = 0$. The remaining expectation is $\langle j, m| J_z |j, m\rangle = m$. Thus the θ coefficient is zero and the ϕ coefficient is $im - im\cos\theta$. Differentiate, integrate over the sphere, and apply the transport formula to get the stated values. The stated matrix data suffice for these rotation and weight identities. Berry, §4, pp. 50–51, obtains the same spin phase using ladder operators. ∎

The quantity $2m$ is an integer, so changing the solid angle by 4π does not change the phase factor. For the ground line of $H = -\kappa \mathbf{B} \cdot \mathbf{J}$ with $\kappa > 0$, the weight is $m = j$ and the degeneracy is again at $\mathbf{B} = 0$.

Self-check. Take $\theta_0 = \arccos(1/3)$ and $\phi(s) = -4\pi s$ for $s \in [0, 1]$. Starting with $c(0) = 1$, compute the ground-state transport factor in both northern and southern frames. Find the signed solid angle for two clockwise traversals of the northern cap and the excited-state phase.

Answer. The azimuth changes by -4π. The northern potential is $id\phi/3$, so the transport equation gives $c(s) = e^{i4\pi s/3}$. The phase and factor are

$$\gamma_N = \frac{4\pi}{3}, \qquad e^{i\gamma_N} = -\frac{1}{2} - \frac{\sqrt{3}}{2} i.$$

The southern potential is $-2id\phi/3$ and gives $\gamma_S = -8\pi/3$. The phases differ by -4π, so their factors agree. The signed solid angle is $-2 \cdot 2\pi(1 - 1/3) = -8\pi/3$, verifying $\gamma = -\Omega/2$. The excited potential has the opposite sign and gives phase $-4\pi/3$ modulo 2π.

Summary. A smooth simple eigenline has the unitary connection $\mathcal{A} = \langle n|d n\rangle$ and curvature $\langle d n| \wedge |d n\rangle$. The cited gapped adiabatic theorem identifies its holonomy with the geometric factor after removing the dynamical phase. The imaginary connection and real potential satisfy $A_B = i\mathcal{A}$. For $H = -\mathbf{B} \cdot \boldsymbol{\sigma}$ with $\mathbf{B} \neq 0$, the ground line has Chern number -1 on an outward direction sphere and $\gamma = -\Omega/2$ modulo 2π. The excluded degeneracy is the source of this parameter-space curvature. Lecture 25 integrates the same curvature over the momentum torus of a gapped band.

Reading. Berry, *Quantal phase factors accompanying adiabatic changes*, §2 and §4, pp. 46–48 and 50–51. Simon, *Holonomy, the quantum adiabatic theorem and Berry's phase*, pp. 2167–2168. Nakahara, §10.6. Tong, *The Quantum Hall Effect*, current §1.5.1–1.5.2, pp. 28–34, with $A_{\text{Tong}} = -A_B$.